Artificial Intelligence for Suturing and Knot-tying Skills Assessment: A Systematic Review.

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Abstract

Artificial intelligence (AI) has great potential for surgical skill training and assessment. However, the heterogeneity of AI models for suturing and knot-tying skills assessment has limited their transformation to educational practice. Our study aimed to explore the utility of AI methods and identify potential challenges for AI-based suturing and knot-tying skills assessment. We searched the PubMed, Web of Science and Embase databases from inception to June 30, 2024, for original studies that adopted AI for suturing and knot-tying skills assessment. Studies that used AI only for gesture or phase recognition, non-English language articles, reviews and conference abstracts were excluded. The data extracted consisted of study characteristics, input data, AI methods and accuracy. Forty-one studies with 807 participants were included. The majority of the studies (n = 20, 49 %) assessed both the suturing and knot-tying skills. In 27 studies (66%), kinematic data such as velocity (40.9 ± 0.9%) were employed for model training with 17 used for robotic surgery, whereas 22 studies (54%) used video data. AI models were used for simple classification in 38 studies (93%), with 26 reporting accuracy (60-100%). The convolutional neural network (CNN) demonstrated more consistent performance (91%-100%) than traditional machine learning (60%-100%). Three studies integrated physiological data with two improved performances (R^2 = 0.92). CNN models trained on spatiotemporal features or multimodal data effectively assessed suturing and knot-tying skills in open, laparoscopic and robotic surgeries, with results highly consistent with traditional scoring. Future studies should focus on standardizing the input data, enhancing cross-scenario transfer, and realizing automated scoring and feedback.

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